Setting up the problem

- In the prior lesson, you learned that:

$$\Sigma_{\widetilde{w}} = \int_0^{\Delta t} e^{A(\Delta t-\tau)} B_w S_w B_w^T e^{A^T(\Delta t-\tau)}\,\mathrm{d}\tau.$$

- But, how to compute this? You learned that if $\Delta t \to 0$, then $\Sigma_{\widetilde{w}} \approx B_w S_w B_w^T$.
- But, how do we find $\Sigma_{\widetilde{w}}$ precisely for general Δt? Assume that we know S_w and plant dynamics.
- Define:

$$Z = \left[\begin{array}{c|c} -A & B_w S_w B_w^T \\ \hline 0 & A^T \end{array}\right], \qquad C = e^{Z\Delta t} = \left[\begin{array}{c|c} c_{11} & c_{12} \\ \hline 0 & c_{22} \end{array}\right].$$

Computing C

- We seek to compute $C = \mathcal{L}^{-1}\left[(sI - Z)^{-1}\right]\Big|_{t=\Delta t}$.
- Then, if $M = sI - Z$, recognize the following matrix identity:

$$M = \left[\begin{array}{c|c} sI - z_{11} & -z_{12} \\ \hline 0 & sI - z_{22} \end{array}\right];\ M^{-1} = \left[\begin{array}{c|c} (sI - z_{11})^{-1} & (sI - z_{11})^{-1} z_{12} (sI - z_{22})^{-1} \\ \hline 0 & (sI - z_{22})^{-1} \end{array}\right].$$

- Then, we have that:

$$\begin{aligned} c_{11} &= \mathcal{L}^{-1}\left[(sI - z_{11})^{-1}\right]\Big|_{t=\Delta t} = \mathcal{L}^{-1}\left[(sI + A)^{-1}\right]\Big|_{t=\Delta t} = e^{-A\Delta t} \\ c_{12} &= \mathcal{L}^{-1}\left[(sI - z_{11})^{-1} z_{12} (sI - z_{22})^{-1}\right]\Big|_{t=\Delta t} \\ &= \mathcal{L}^{-1}\left[(sI + A)^{-1} B_w S_w B_w^T (sI - A^T)^{-1}\right]\Big|_{t=\Delta t} \\ c_{22} &= \mathcal{L}^{-1}\left[(sI - z_{22})^{-1}\right]\Big|_{t=\Delta t} = \mathcal{L}^{-1}\left[(sI - A^T)^{-1}\right]\Big|_{t=\Delta t} = e^{A^T\Delta t} = A_d^T. \end{aligned}$$

The usefulness of C

- So, $c_{22}^T = A_d$. Also, we will prove that $c_{22}^T c_{12} = \Sigma_{\widetilde{w}}$.
- To show this, first recognize that the transfer function of a state-space model of a system a, b, c ($d = 0$) is equal to $H(s) = c(sI - a)^{-1}b$.
  - So, $(sI + A)^{-1} B_w S_w B_w^T (sI - A^T)^{-1}$ represents the cascade of two systems.
    - One has: $c = I$, $a = -A$, $b = B_w$; other has: $c = S_w B_w^T$, $a = A^T$, $b = I$.
- The term c_{12} is the inverse Laplace transform of this cascade, which is the same as the impulse response of the cascade.
- The impulse response of the cascade is the convolution of the impulse responses of the two individual systems comprising the cascade.
- Note that the impulse response of a state-space model ($d = 0$) is $ce^{at}b$.
  - So, the two systems have impulse responses $Ie^{-At}B_w$ and $S_w B_w^T e^{A^T t}$.

Computing $\Sigma_{\widetilde{w}}$

- The impulse response of the whole is the convolution of the two cascaded impulse responses:

$$\int_0^{\Delta t} I A^{-\tau} B_w S_w B_w^T e^{A^T(\Delta t - \tau)}\, \mathrm{d}\tau.$$

- Multiplying this result on the left by $c_{22}^T = A_d = e^{A\Delta t}$,

$$c_{22}^T c_{12} = \int_0^{\Delta t} e^{A(\Delta t - \tau)} B_w S_w B_w^T e^{A^T(\Delta t - \tau)}\, \mathrm{d}\tau = \Sigma_{\widetilde{w}}.$$

- In conclusion, with one `expm.m` command, we can quickly convert dynamics from continuous-time to discrete-time:

$$\boxed{A_d = c_{22}^T \qquad \text{and} \qquad \Sigma_{\widetilde{w}} = c_{22}^T c_{12}.}$$

A worked example (1)

- Let:

$$\dot{x}(t) = \underbrace{\begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}}_{A} x(t) + \underbrace{\begin{bmatrix} 0 \\ 1 \end{bmatrix}}_{B_w} w(t)$$

with $S_w = 0.06$ and $\Delta t = 2\pi/16$. Then, $B_w S_w B_w^T = \begin{bmatrix} 0 & 0 \\ 0 & 0.06 \end{bmatrix}$ and:

$$Z = \left[\begin{array}{cc|cc} 0 & -1 & 0 & 0 \\ 1 & 1 & 0 & 0.06 \\ \hline 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & -1 \end{array}\right]; \qquad C = \left[\begin{array}{cc|cc} 0.91 & -0.47 & -0.006 & -0.0046 \\ 0.47 & 1.38 & 0.0046 & 0.023 \\ \hline 0 & 0 & 0.93 & -0.32 \\ 0 & 0 & 0.32 & 0.62 \end{array}\right].$$

A worked example (2)

- From c_{12} and c_{22}, we can compute the following quantities very quickly:

$$A_d = \begin{bmatrix} 0.93 & 0.32 \\ -0.32 & 0.62 \end{bmatrix}; \; \Sigma_{\widetilde{w}} = \begin{bmatrix} 0.0009 & 0.0030 \\ 0.0030 & 0.0156 \end{bmatrix}; \; \text{also, } B_w S_w B_w^T = \begin{bmatrix} 0 & 0 \\ 0 & 0.06 \end{bmatrix}.$$

- Exact $\Sigma_{\widetilde{w}}$ and approximate $\Sigma_{\widetilde{w}}$ are very different due to large Δt.
- Compare predictions of steady-state performance.
  - Continuous: $A\Sigma_{\tilde{x},ss} + \Sigma_{\tilde{x},ss}A^T + B_w S_w B_w^T = 0;\quad \Sigma_{\tilde{x},ss} = \begin{bmatrix} 0.03 & 0 \\ 0 & 0.03 \end{bmatrix}.$
  - Discrete: $\Sigma_{\tilde{x},ss} = A_d \Sigma_{\tilde{x},ss} A_d^T + \Sigma_{\widetilde{w}};\qquad \Sigma_{\tilde{x},ss} = \begin{bmatrix} 0.03 & 0 \\ 0 & 0.03 \end{bmatrix}.$
- Because we used discrete white noise scaled to be equivalent to the continuous noise, the steady-state predictions are the same.

What about converting S_v to $\Sigma_{\tilde{v}}$

- This is probably a good time to introduce converting S_v to $\Sigma_{\tilde{v}}$.
- In continuous-time, the sensor noise $v(t)$ is assumed to be zero mean and have $\mathbb{E}[v(t_1)v(t_2)^T] = S_v\delta(t_1 - t_2)$ where $S_v > 0$.
- Approach to converting to discrete time: analyze the discrete case as $\Delta t \to 0$:

$$v_k = \frac{1}{\Delta t}\int_0^{\Delta t} v(t)\,\mathrm{d}t \qquad \text{(average value)}.$$

- Then,

$$\Sigma_{\tilde{v}} = \mathbb{E}[v_k v_k^T] = \frac{1}{\Delta t^2}\int_0^{\Delta t}\int_0^{\Delta t} \mathbb{E}[v(\gamma)v(\tau)^T]\,\mathrm{d}\gamma\,\mathrm{d}\tau = \frac{1}{\Delta t^2}\cdot \Delta t \cdot S_v = \frac{S_v}{\Delta t}.$$

- As $\Delta t \to 0$, the discrete noise tends to infinite impulse-like term having weight S_v, similar to $S_v\delta(t)$.

Summary

- The conversion between continuous-time and discrete-time is now complete:

DISCRETE: $A_d,\ \Sigma_{\widetilde{w}},\ B_d$;

$$\begin{aligned}\text{Given } \Sigma_{\tilde{x},0} &: \quad \Sigma_{\tilde{x},k} = A_d\Sigma_{\tilde{x},k-1}A_d^T + \Sigma_{\widetilde{w}}\\ \text{Given } \bar{x}_0 &: \quad \bar{x}_k = A_d\bar{x}_{k-1} + B_d u_{k-1}.\end{aligned}$$

CONTINUOUS: $A,\ S_w,\ B_u,\ B_w$;

$$\begin{aligned}\text{Given } \Sigma_{\tilde{x}}(0) &: \quad \dot{\Sigma}_{\tilde{x}}(t) = A\Sigma_{\tilde{x}}(t) + \Sigma_{\tilde{x}}(t)A^T + B_w S_w B_w^T\\ \text{Given } \bar{x}(0) &: \quad \dot{\bar{x}}(t) = A\bar{x}(t) + B_u u(t).\end{aligned}$$

- All matrices may be time varying.
- Equivalent $\Sigma_{\widetilde{w}}$ for given S_w can be computed using the trick of this lesson.
- Equivalent $\Sigma_{\tilde{v}}$ for given S_v can also be computed as $\Sigma_{\tilde{v}} = S_v/\Delta t$.